HOSPITAL MANAGEMENT SYSTEM



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CASE

OUR ENTREPRENEURIAL TEAM IS LAUNCHING A SPECIALIZED HOSPITAL IN ALANYA TO ADDRESS THE LIMITATIONS OF THE CURRENT PUBLIC HEALTHCARE SYSTEM, ESPECIALLY THE LACK OF DOCTORS, OUTDATED INFRASTRUCTURE, AND SEASONAL PATIENT OVERLOAD.

UNLIKE GENERAL HOSPITALS, OUR MODEL FOCUSES ON TWO HIGH-DEMAND DEPARTMENTS: INTERNAL MEDICINE AND ORTHOPEDICS. THE HOSPITAL OPERATES THREE DAYS A WEEK (MONDAY, WEDNESDAY, AND FRIDAY) FROM 9:00 AM TO 4:00 PM, WITH APPOINTMENTS ENDING AT 3:00 PM. EACH DOCTOR SEES A MAXIMUM OF 7 PATIENTS PER DAY, DEDICATING A FULL HOUR PER CONSULTATION FOR BETTER QUALITY CARE AND REDUCED FATIGUE.

PATIENTS BOOK APPOINTMENTS ONLINE BASED ON DOCTOR AND DEPARTMENT AVAILABILITY, LIMITED TO ONE APPOINTMENT PER WEEK TO MAXIMIZE ACCESS FOR OTHERS. DOCTORS CAN VIEW BUT NOT MODIFY THEIR SCHEDULES AND PATIENT DETAILS; THEY CAN CONFIRM APPOINTMENTS AND ISSUE PRESCRIPTIONS THROUGH THE SYSTEM.

PRESCRIPTIONS ARE STORED DIGITALLY AND ACCESSED BY PATIENTS VIA ID AT PHARMACIES. OUR SYSTEM ENSURES EFFICIENT SCHEDULING, SECURE DATA HANDLING, AND A FOCUSED, HIGH-QUALITY CARE EXPERIENCE TAILORED TO LOCAL HEALTHCARE CHALLENGES.

FUNCTIONAL REQUIREMENTS

FR-01: Users must be able to log in with their credentials.
FR-02: Users must be able to create a new account.
FR-03: Patients must be able to select available slots and schedule appointments.
FR-04: The system must restrict scheduling to hospital operating days (Mon, Wed, Fri) and hours (9:00 AM - 4:00 PM).
FR-05: The system must block appointments after 3:00 PM to end by 4:00 PM.
FR-06: The system must limit each doctor to 7 patients per day.
FR-07: The system must restrict patients to one appointment per week.
FR-08: The system must send booking confirmation notifications to patients.
FR-09: Doctors must be able to view their daily and weekly schedules.
FR-10: Doctors must be able to access patient details for appointments.
FR-11: Doctors must be able to confirm appointments without modifying them.
FR-12: Doctors must be able to write and save prescriptions post-consultation.
FR-13: The system must store prescriptions securely, linked to patient IDs.
FR-14: Patients must be able to retrieve prescriptions from pharmacies using their ID.
FR-15: Admins must be able to manage user account.
FR-16: Admins must be able to configure scheduling rules.

NON-FUNCTIONAL REQUIREMENTS

NFR-01: The system must update appointment availability in real time.

NFR-02: The system must support multiple users without delays.

NFR-03: The system must use secure authentication and encryption.

NFR-04: The system must comply with healthcare data laws (e.g., HIPAA, GDPR).

NFR-05: The system must have an easy-to-use interface.

NFR-06: The system must maintain 99.9% uptime during operating hours.

NFR-07: The system must prevent double-booking and errors.

NFR-08: The system must scale to support more departments and doctors.

NFR-09: The system must be maintainable with clear code and documentation.

NFR-10: The system must restrict prescription access to authorized users.

ENTITIES

1. Appointment Management Module
- Book, reschedule, and cancel appointments.
- Display available slots based on doctor's schedule.
- Send notifications and reminders to patients and doctors.

2. Doctor Management Module
- Define working hours and availability.
- Assign doctors to specific polyclinics.

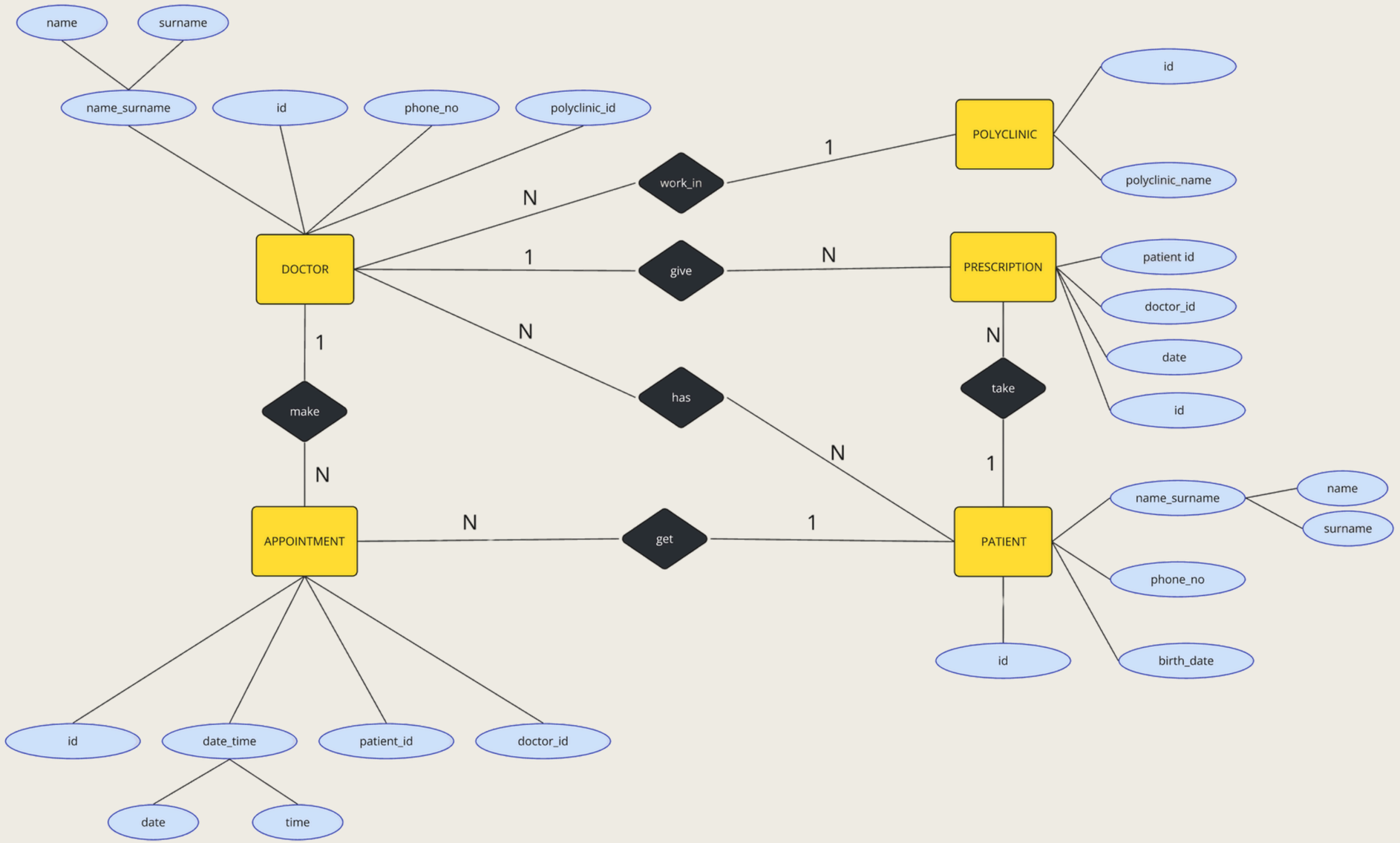
3.Patient Management Module
- Store patient details including contact information and birth date.

4. Prescription Management Module
- Allow doctors to create and issue prescriptions.
- Enable patients to view their prescriptions.

5.Polyclinic Management Module
- Manage polyclinic details and doctors assigned to them.



ER DIAGRAM


name
surname
name_surname
id
phone_no
polyclinic_id
DOCTOR
work_in
1
N
POLYCLINIC
id
polyclinic_name
1
give
N
PRESCRIPTION
patient id
doctor_id
date
id
1
make
N
N
has
N
take
1
APPOINTMENT
N
get
1
PATIENT
name_surname
name
surname
phone_no
birth_date
id
id
date_time
patient_id
doctor_id
date
time

CREATION OF A DATABASE AND TABLES

```sql
CREATE DATABASE hospital_management_system;
USE hospital_management_system;


CREATE TABLE polyclinic (
id INT Not Null PRIMARY KEY,
polyclinic_name varchar(100) Not Null
);

CREATE TABLE doctor (
id INT Not Null PRIMARY KEY,
name_surname varchar(100) Not Null,
polyclinic_id INT Not Null,
phone_no varchar(100) Not Null,
FOREIGN KEY(polyclinic_id) references polyclinic(id) ON DELETE restrict ON UPDATE restrict
);

CREATE TABLE patient (
id INT Not Null PRIMARY KEY,
name_surname varchar(100) Not Null,
birth_date date Not Null,
phone_no varchar(100) Not Null
);
```

```
25  CREATE TABLE appointment (
26  id INT Not Null PRIMARY KEY,
27  doctor_id INT Not Null,
28  patient_id INT Not Null,
29  date_time datetime Not Null,
30  FOREIGN KEY(doctor_id) references doctor(id) ON DELETE cascade ON UPDATE cascade,
31  FOREIGN KEY(patient_id) references patient(id) ON DELETE cascade ON UPDATE cascade
32  );
33
34  CREATE TABLE prescription (
35  id INT Not Null PRIMARY KEY,
36  patient_id INT Not Null,
37  doctor_id INT Not Null,
38  date date Not Null,
39  FOREIGN KEY(patient_id) references patient(id) ON DELETE cascade ON UPDATE cascade,
40  FOREIGN KEY(doctor_id) references doctor(id) ON DELETE cascade ON UPDATE cascade
41  );
```

ADDING ATTRIBUTE INFORMATION TO TABLES

```sql
USE hospital_management_system;

INSERT INTO polyclinic (id, polyclinic_name) VALUES
(1, "Internal Medicine Polyclinic"),
(2, "Orthopedics Polyclinic");


INSERT INTO doctor (id, name_surname, polyclinic_id, phone_no) VALUES
(1, "Dr.Mert Gülsuyu", 1, "555-444-3322"),
(2, "Dr. Yıldız Kemal", 2, "522-556-8972"),
(3, "Dr. Zahide Yetiş", 2, "555- 235-6689"),
(4, "Dr. Orhan Gencebay", 1, "522-189-3232"),
(5, "Dr. Canan Karatay", 1, "555-198-2998");


INSERT INTO patient (id, name_surname, birth_date, phone_no) VALUES
(1, 'Ali Yılmaz', '1985-03-02', '532-123-4567'),
(2, 'Ayşe Demir', '1990-07-14', '542-345-6789'),
(3, 'Mehmet Kaya', '1983-11-23', '551-456-7890'),
(4, 'Zeynep Çelik', '1978-05-07', '533-567-8901'),
(5, 'Hasan Şahin', '1995-12-12', '541-678-9012'),
(6, 'Emine Özkan', '1988-06-30', '553-789-0123'),
(7, 'Ahmet Karaca', '1981-09-19', '535-890-1234'),
(8, 'Fatma Eren', '1992-04-04', '544-901-2345'),
(9, 'Yusuf Aydın', '1987-10-16', '555-012-3456'),
(10, 'Elif Doğan', '1994-02-25', '531-123-4567'),
```

- We enter the information in this way for all 105 patients.

```sql
INSERT INTO prescriptions (id, patient_id, doctor_id, date) VALUES
(314, 27, 4, '2025-05-30'),
(347, 18, 2, '2025-05-26'),
(363, 102, 5, '2025-05-28'),
(385, 99, 1, '2025-05-26'),
(328, 4, 3, '2025-05-26'),
(332, 101, 3, '2025-05-26'),
(336, 64, 5, '2025-05-26'),
(206, 1, 4, '2025-05-30'),
(331, 60, 2, '2025-05-28'),
(232, 19, 1, '2025-05-28'),
(239, 82, 1, '2025-05-30'),
(333, 67, 5, '2025-05-28'),
(277, 6, 3, '2025-05-26'),
(318, 50, 4, '2025-05-28'),
(383, 92, 2, '2025-05-26'),
(218, 95, 3, '2025-05-26'),
(220, 20, 2, '2025-05-26'),
```

- We enter the information in this way for all 105 prescriptions.

```sql
INSERT INTO appointment (id, patient_id, doctor_id, date_time) VALUES
(1492, 10, 4, '2025-05-30 10:00:00'),
(1574, 36, 2, '2025-05-26 10:00:00'),
(1948, 30, 5, '2025-05-28 13:00:00'),
(1787, 83, 1, '2025-05-26 11:00:00'),
(1798, 33, 3, '2025-05-26 14:00:00'),
(1799, 9, 3, '2025-05-26 09:00:00'),
(1472, 105, 5, '2025-05-26 15:00:00'),
(1392, 37, 4, '2025-05-30 14:00:00'),
(1493, 40, 2, '2025-05-28 11:00:00'),
(1212, 89, 1, '2025-05-28 09:00:00'),
(1478, 34, 1, '2025-05-30 12:00:00'),
(1125, 103, 5, '2025-05-28 15:00:00'),
(1246, 31, 3, '2025-05-26 11:00:00'),
(1775, 74, 4, '2025-05-28 14:00:00'),
(1243, 48, 2, '2025-05-26 15:00:00'),
(1932, 84, 3, '2025-05-26 12:00:00'),
(1094, 32, 2, '2025-05-26 13:00:00'),
```

- We enter the information in this way for all 105 appointments.

QUERYING THE INFORMATION WE WANT TO SEE

```sql
USE hospital_management_system;


SELECT name_surname, phone_no FROM patient;
-- Show only patient names and telephone numbers --

SELECT * FROM patient LIMIT 10;
-- List the top 10 patients --

SELECT * FROM doctor WHERE id = 5;
-- Find a specific doctor by ID = 5 --

SELECT * FROM patient ORDER BY name_surname;
-- Order from the patient with the most recent date of birth to the oldest --
```

```sql
SELECT
    a.id AS appointment_id,
    a.date_time AS appointment_date,
    d.name_surname AS doctor_name,
    pc.polyclinic_name AS poliyclinic
FROM appointment a
JOIN doctor d ON a.doctor_id = d.id
JOIN polyclinic pc ON d.polyclinic_id = pc.id
WHERE a.patient_id = 10
ORDER BY a.date_time;
--  See all appointment information for a patient_id = 10 --
```

```sql
SELECT
    a.id AS appointment_id,
    a.date_time AS appointment_date_time,
    p.id AS patient_id,
    p.name_surname AS patient_name,
    p.birth_date AS patient_birth_date,
    p.phone_no AS patient_phone,
    d.name_surname AS doctor_name,
    pc.polyclinic_name AS poliyclinic_name
FROM appointment a
JOIN patient p ON a.patient_id = p.id
JOIN doctor d ON a.doctor_id = d.id
JOIN polyclinic pc ON d.polyclinic_id = pc.id
WHERE a.doctor_id = 4 AND DATE(a.date_time) = '2025-05-28'
ORDER BY a.date_time;
    -- Appointment and patient information of the doctor with doctor_id = 4 on '2025-05-28' --
```

```sql
SELECT
    a.id AS appointment_id,
    a.date_time AS appointment_date,
    d.name_surname AS doctor_name,
    pc.polyclinic_name AS polyclinic
FROM appointment a
JOIN doctor d ON a.doctor_id = d.id
JOIN polyclinic pc ON d.polyclinic_id = pc.id
JOIN patient p ON a.patient_id = p.id
WHERE p.name_surname LIKE '%Ahmet%'
ORDER BY a.date_time;
-- List Appointments by Patient Name = Ahmet --

SELECT
    d.id AS doctor_id,
    d.name_surname AS doctor_name,
    d.phone_no AS doctor_phone,
    p.id AS poliyclinic_id,
    p.polyclinic_name AS poliyclinic_adi
FROM doctor d
JOIN polyclinic p ON d.polyclinic_id = p.id
WHERE p.id = 2
ORDER BY d.name_surname;

    -- Listing doctors, doctors' phone number and polyclinic's name with polyclinic_id = 2 --
```

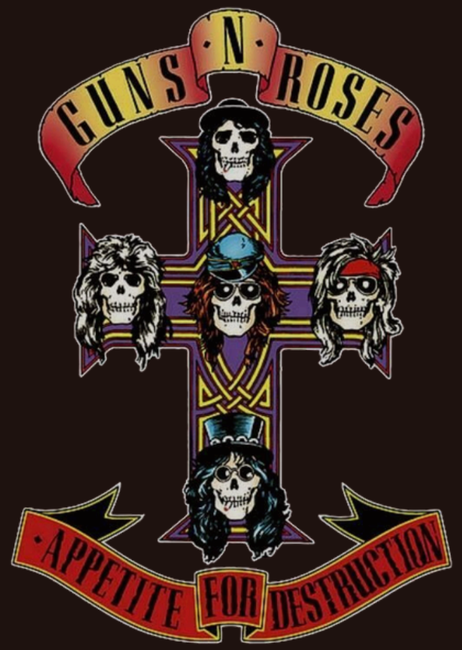
```sql
SELECT
    a.id AS appointment_id,
    a.date_time AS appointment_date_time,
    p.id AS patient_id,
    p.name_surname AS patient_name,
    p.birth_date AS patient_birth_date,
    p.phone_no AS patient_phone,
    d.name_surname AS doctor_name,
    pc.polyclinic_name AS poliyclinic_name
FROM appointment a
JOIN patient p ON a.patient_id = p.id
JOIN doctor d ON a.doctor_id = d.id
JOIN polyclinic pc ON d.polyclinic_id = pc.id
WHERE a.doctor_id = 4 AND DATE(a.date_time) = '2025-05-28'
ORDER BY a.date_time;
    -- Appointment and patient information of the doctor with doctor_id = 4 on '2025-05-28' --
```

```sql
SELECT
    a.id AS appointment_id,
    TIME(a.date_time) AS appointment_time,
    p.id AS patient_id,
    p.name_surname AS patient_name,
    p.birth_date AS patient_birth_date,
    p.phone_no AS patient_phone,
    d.id AS doctor_id,
    d.name_surname AS doctor_name,
    pc.polyclinic_name AS polyclinic_name
FROM appointment a
JOIN patient p ON a.patient_id = p.id
JOIN doctor d ON a.doctor_id = d.id
JOIN polyclinic pc ON d.polyclinic_id = pc.id
WHERE TIME(a.date_time) = '14:00:00'
ORDER BY a.date_time;

    -- List doctor and patient information for appointments at 14:00:00 --
```

```sql
SELECT pc.polyclinic_name, COUNT(DISTINCT a.patient_id) AS patient_count
FROM polyclinic pc
JOIN doctor d ON pc.id = d.polyclinic_id
JOIN appointment a ON d.id = a.doctor_id
GROUP BY pc.id, pc.polyclinic_name
ORDER BY patient_count DESC;
    -- Number of patients in each polyclinic --

SELECT DISTINCT
p.id AS patient_id,
p.name_surname AS patient_name,
p.birth_date AS birth_date,
p.phone_no AS phone
FROM patient p
JOIN appointment a ON p.id = a.patient_id
JOIN doctor d ON a.doctor_id = d.id
WHERE d.polyclinic_id = 1
ORDER BY p.name_surname;
    -- List information of patients with polyclinic_id = 1 --
```



THANK YOU